

Corrected JPAR Ranking Diagnostics

Input: data/data_jpar_v2/source_of_truth_calculation_df.csv

Rows analyzed: 3,978

Events analyzed: 51

Ranked members: 1,224

Date range: 2025-04-04 to 2026-03-29

Systems compared: current JPAR, raw JPAR update, raw/log z-scores, percentiles, normalized ranks, Elo, TrueSkill, and conservative blends.

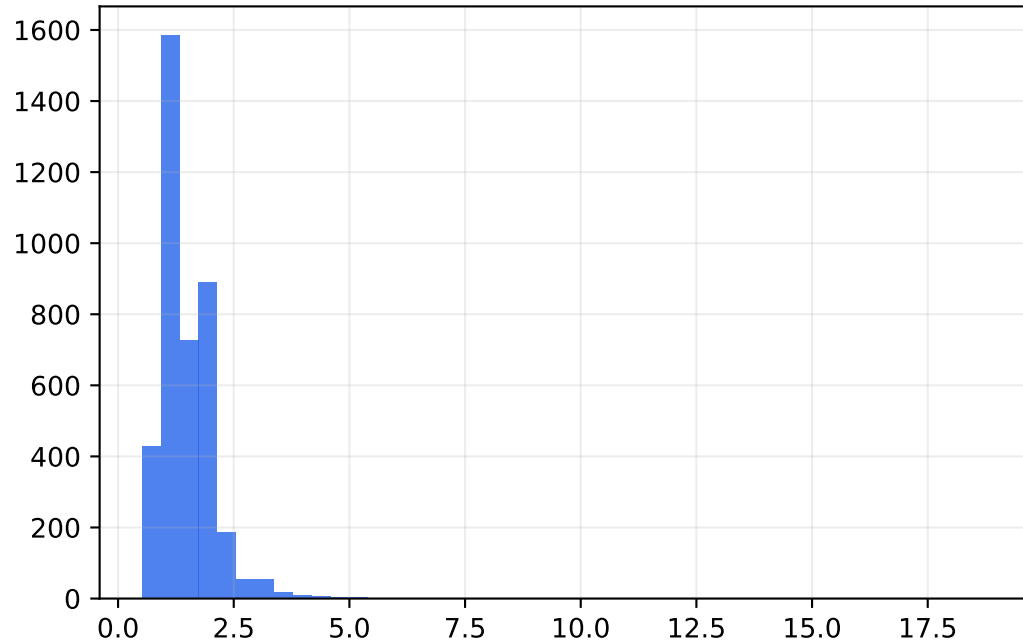
The goal is diagnostic: identify drift, rank disagreements, one-event sensitivity, and event-calibration effects.

Current JPAR Top 25 With Selected Alternative Ranks

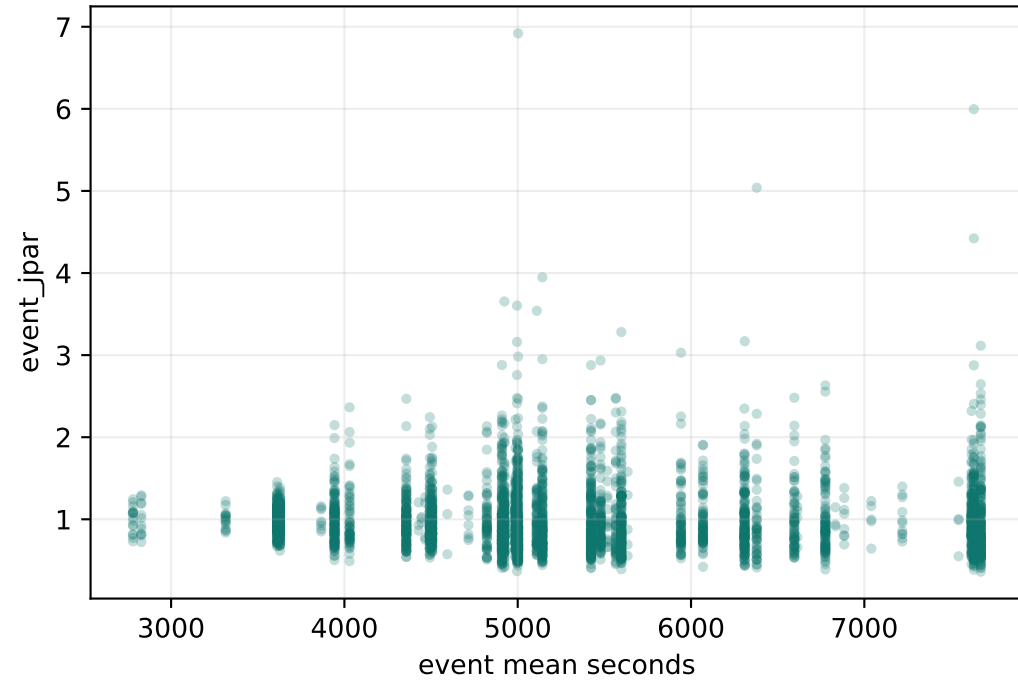
jpar_rank	full_name	events	mean_time	jpar	weighted_log_zs	elo_rating_rank	trueskill_conservation
1	Becca Taylor	1	00:35:00	0.517494	2	167	83
2	Abby Patnoe	2	00:55:34	0.566007	38	110	67
3	Michaela Keener	2	00:48:43	0.618773	6	98	29
4	Alice Rowe	6	00:39:15	0.62667	1	15	1
5	Sam Zucker	2	01:02:42	0.629047	80	142	131
6	Emilie Hoepker	1	01:03:40	0.698703	103	151	295
7	Grayson Shann	7	00:44:22	0.709497	5	13	4
8	Hannah Doyle	9	00:40:39	0.714141	3	4	2
9	Cathy Roiter	6	00:47:36	0.720663	10	24	11
10	Hanna Sigler	7	00:41:51	0.721381	8	14	6
11	Amber Whitmill	3	00:45:12	0.735744	15	72	23
12	Sarah Rowe	2	00:49:39	0.737052	53	123	100
13	Kylie Ledingham	4	00:43:50	0.740631	16	52	21
14	Caryn Knill	12	00:46:54	0.746043	18	2	7
15	Nathan Biberdorf	6	00:45:54	0.748566	13	33	18
16	Jessa Douglas	6	00:48:46	0.749121	17	26	16
17	Tiffany Medeiros	5	00:46:20	0.749226	11	30	8
18	Melinda Shokler	3	00:43:56	0.750373	9	76	34
19	Shea Burkett	5	00:43:22	0.753213	27	46	12
20	Heather Ross	1	01:36:22	0.75365	252	357	467
21	Emma Williams	8	00:46:36	0.754481	23	11	9
22	Allie Arendell	8	00:49:59	0.75486	25	19	31
23	Sarah Stubeck	4	00:45:22	0.756217	29	109	33
24	Yasmeen Almog	2	01:15:09	0.768432	618	422	308
25	Jeanne Roiter	5	00:47:51	0.771096	19	35	15

Core Score And Time Diagnostics

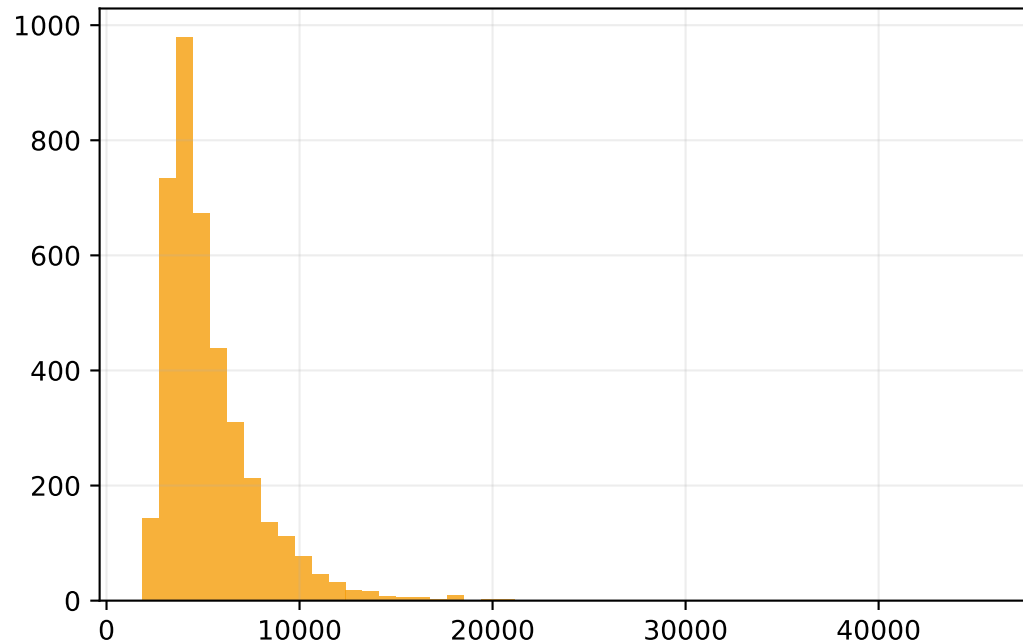
Latest JPAR Distribution



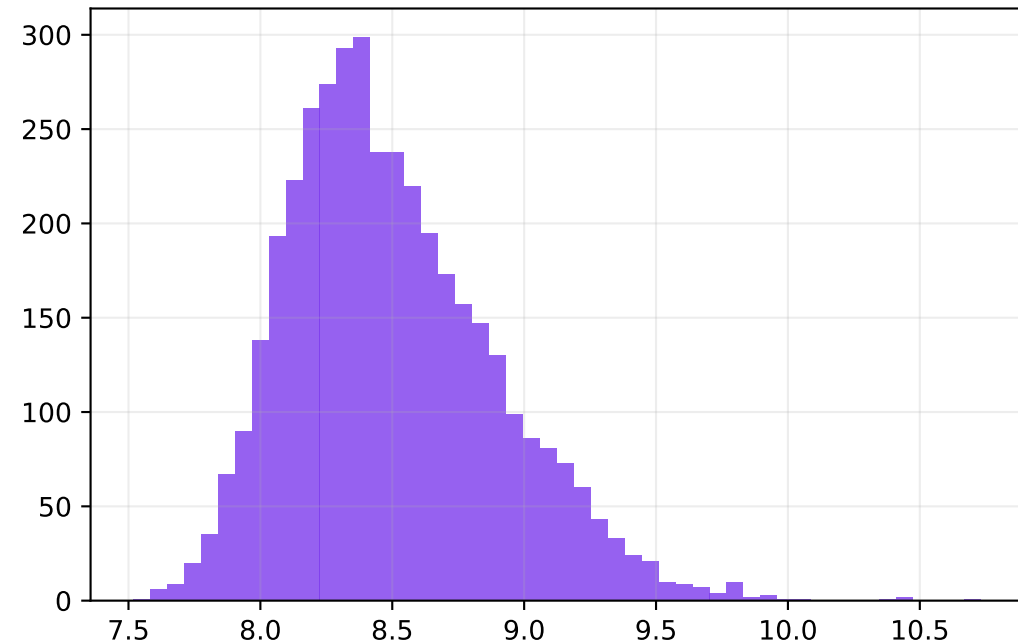
Event Mean Time vs Raw Event JPAR



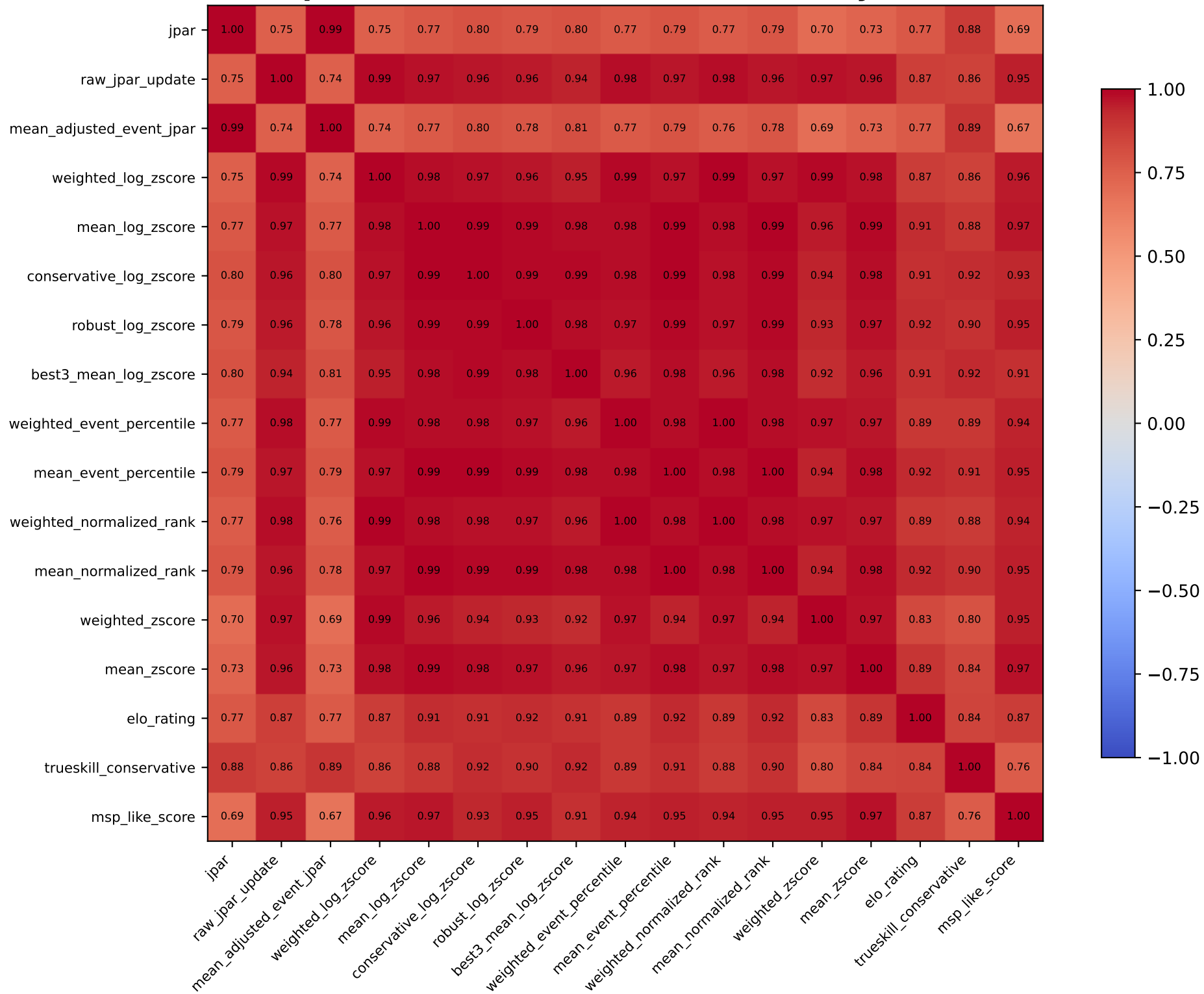
Raw Completion Seconds



Log Completion Seconds

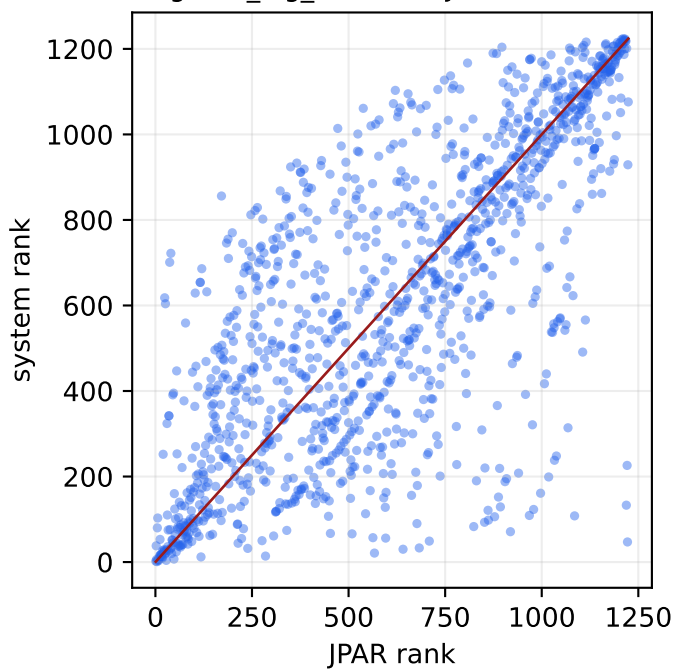


Spearman Rank Correlation Across Systems

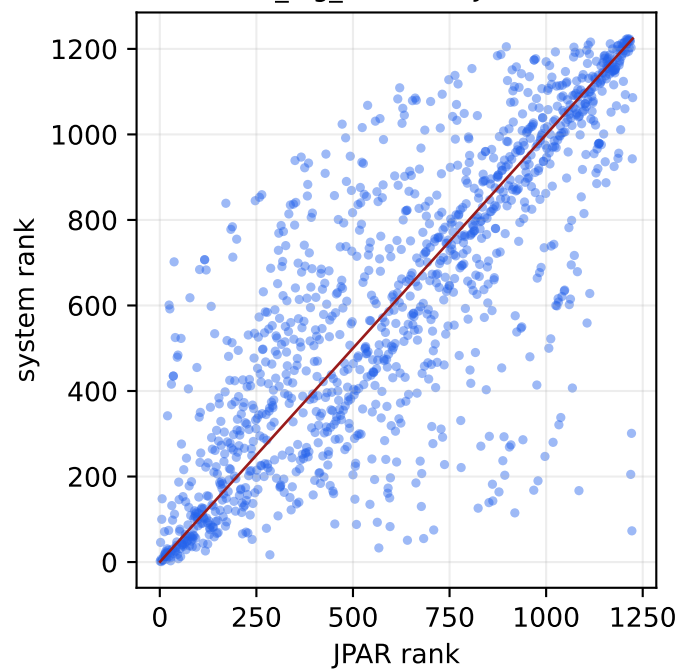


Selected Alternative Ranks vs JPAR

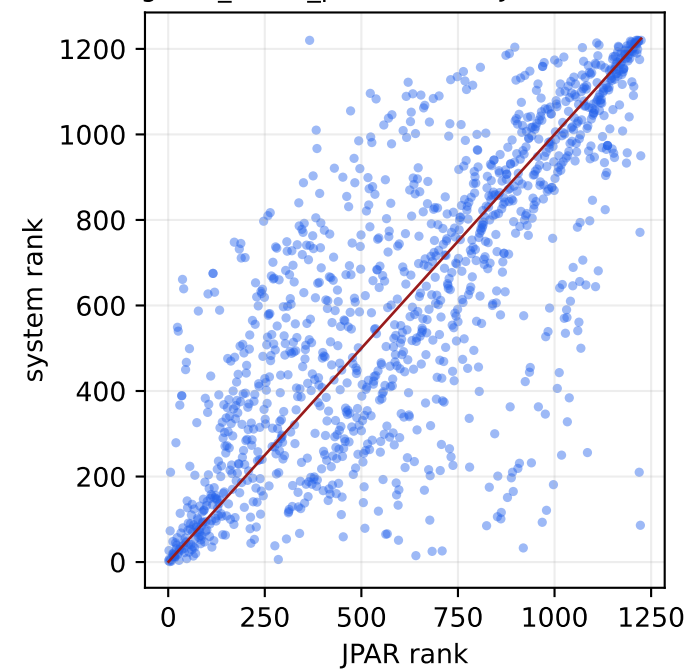
weighted_log_zscore vs JPAR ($\rho=0.75$)



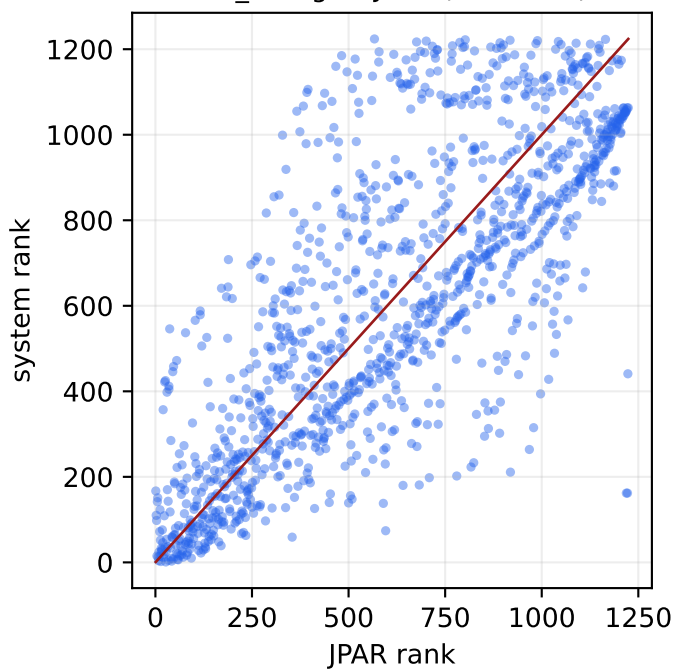
conservative_log_zscore vs JPAR ($\rho=0.80$)



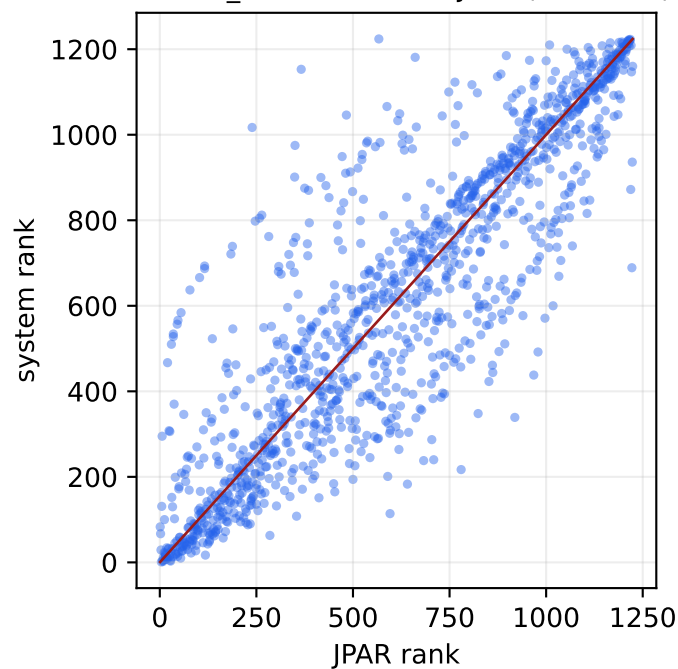
weighted_event_percentile vs JPAR ($\rho=0.77$)



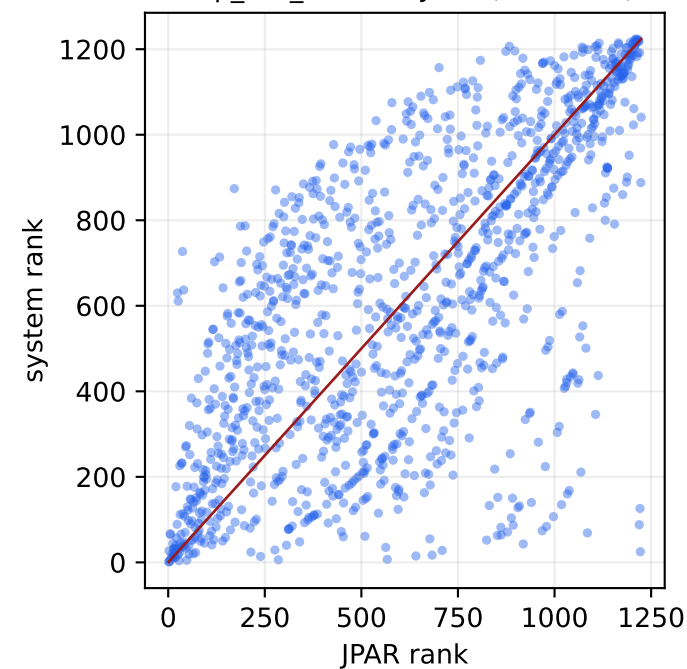
elo_rating vs JPAR ($\rho=0.77$)



trueskill_conservative vs JPAR ($\rho=0.88$)

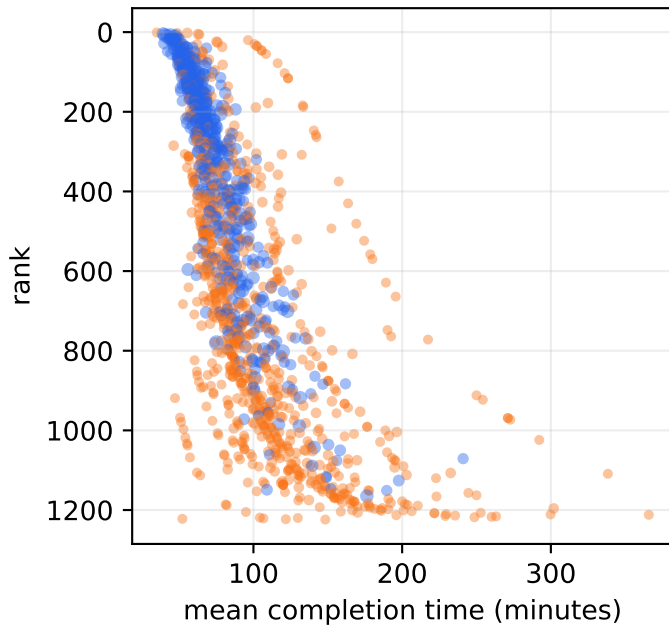


mvp_like_score vs JPAR ($\rho=0.69$)

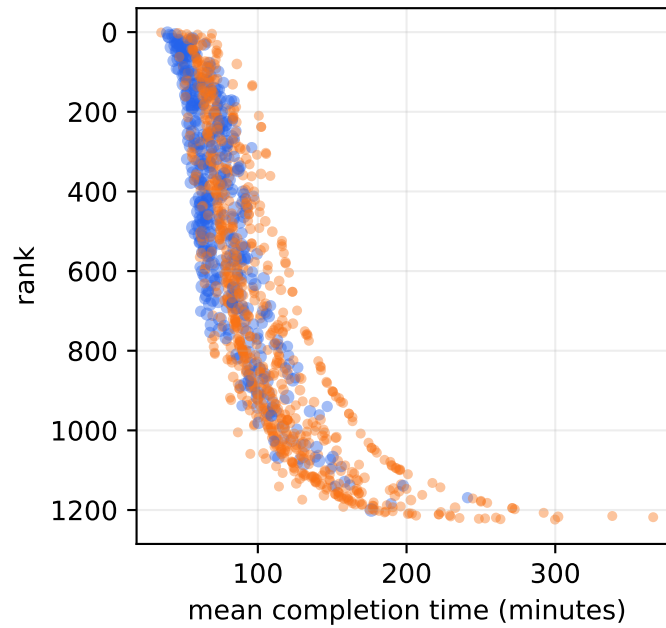


Rank vs Mean Time by Person (1/3)

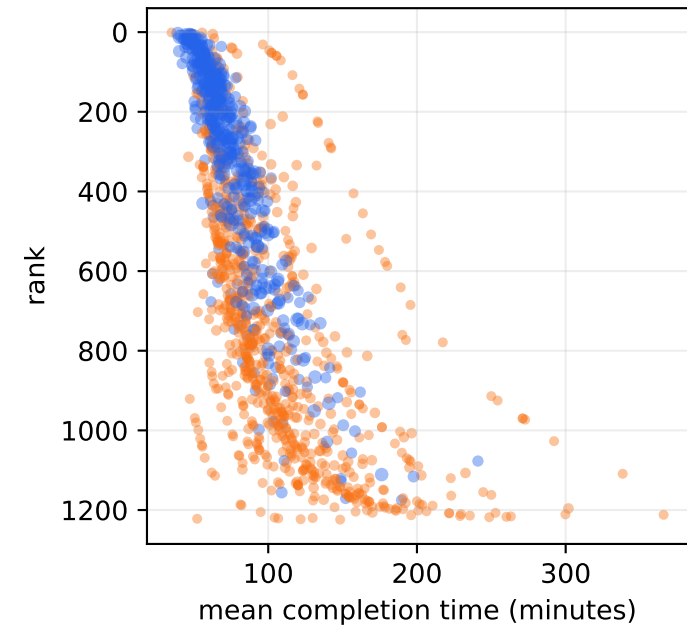
jpar rank vs mean time
 $\rho=0.76$; orange=<3 events



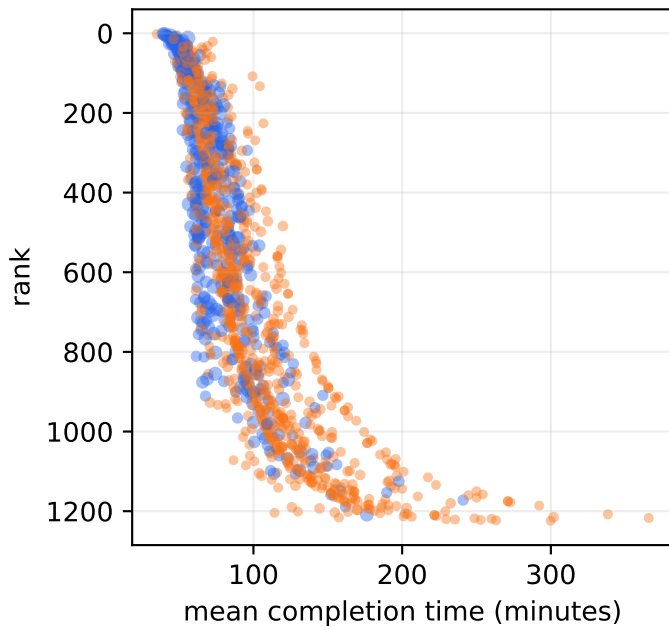
raw_jpar_update rank vs mean time
 $\rho=0.87$; orange=<3 events



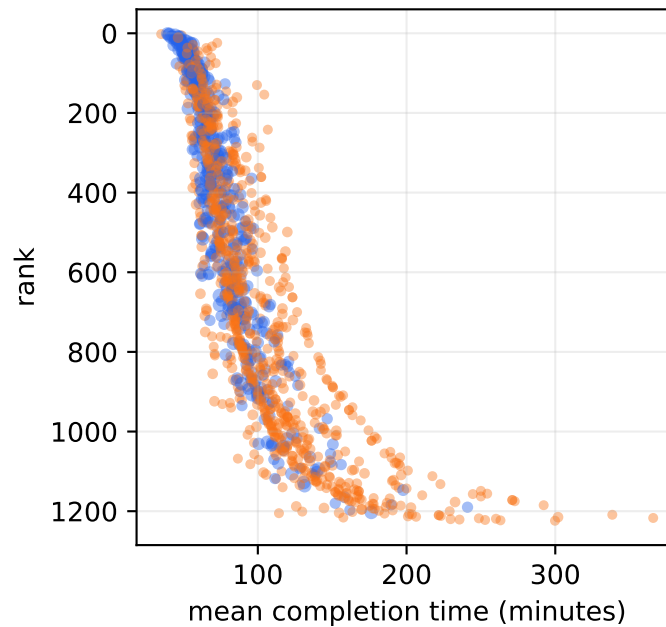
mean_adjusted_event_jpar rank vs mean time
 $\rho=0.75$; orange=<3 events



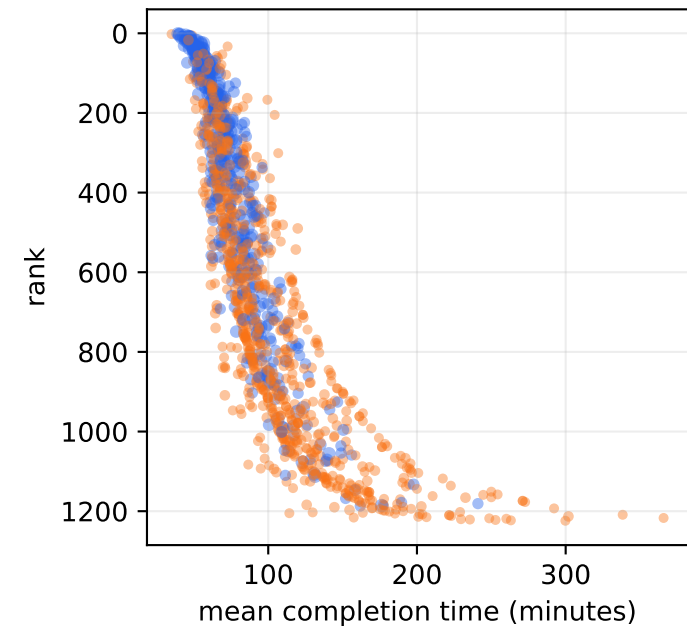
weighted_log_zscore rank vs mean time
 $\rho=0.87$; orange=<3 events



mean_log_zscore rank vs mean time
 $\rho=0.90$; orange=<3 events

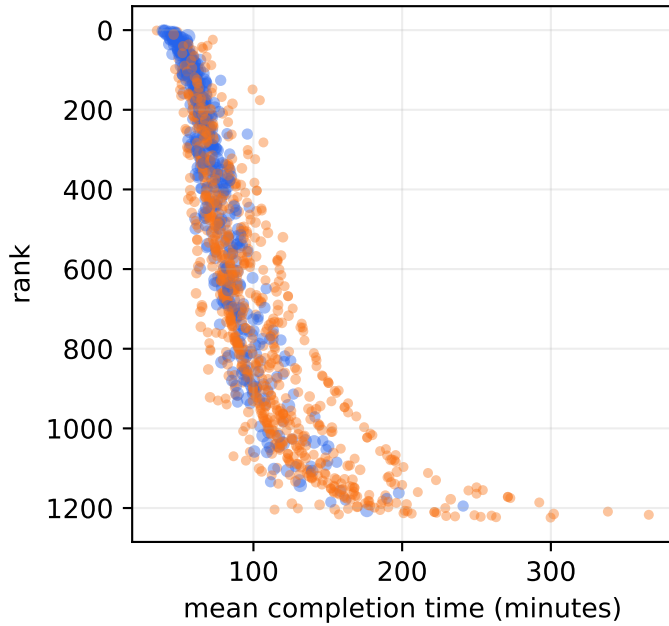


conservative_log_zscore rank vs mean time
 $\rho=0.90$; orange=<3 events

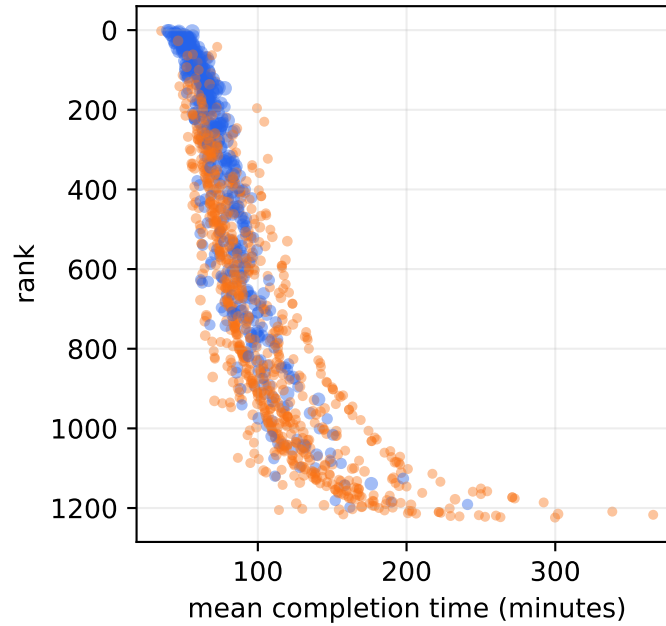


Rank vs Mean Time by Person (2/3)

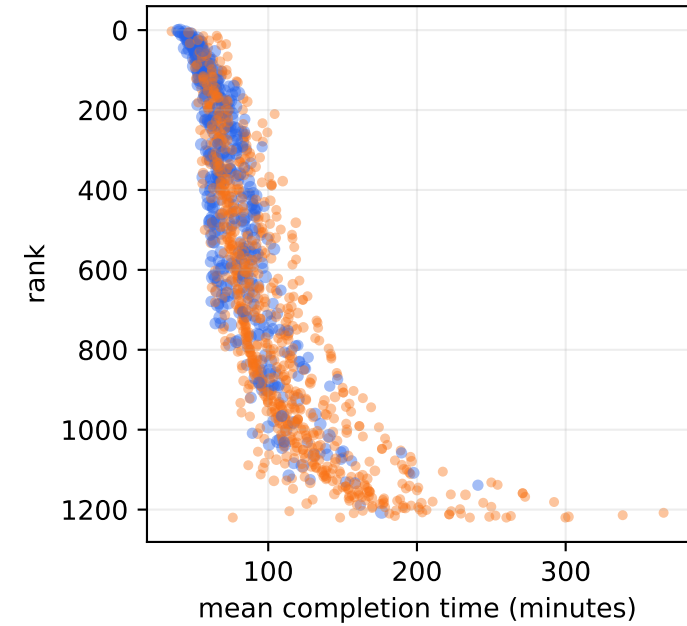
robust_log_zscore rank vs mean time
 $\rho=0.91$; orange=<3 events



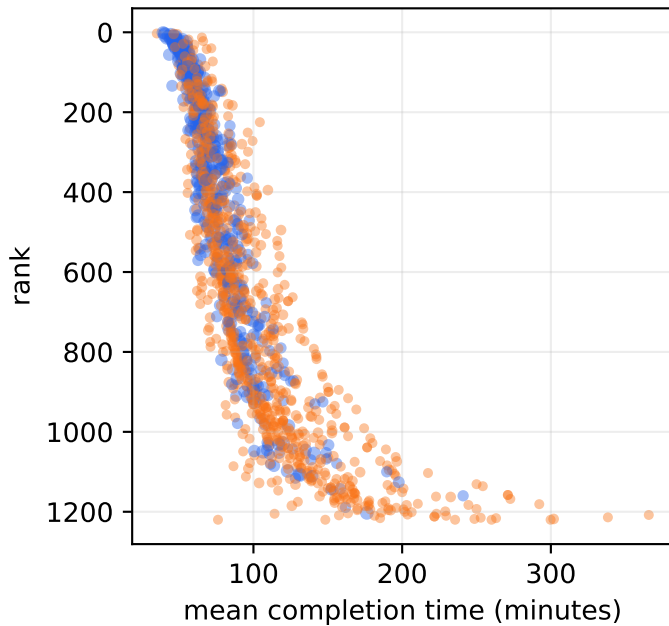
best3_mean_log_zscore rank vs mean time
 $\rho=0.90$; orange=<3 events



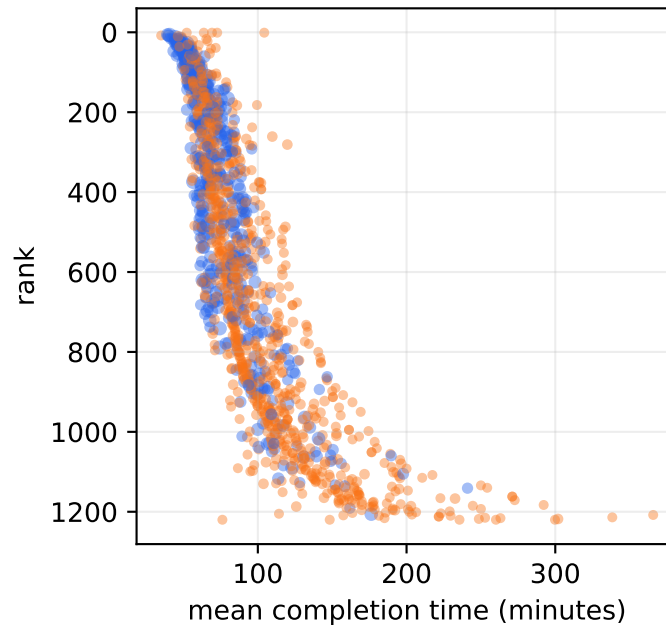
weighted_event_percentile rank vs mean time
 $\rho=0.88$; orange=<3 events



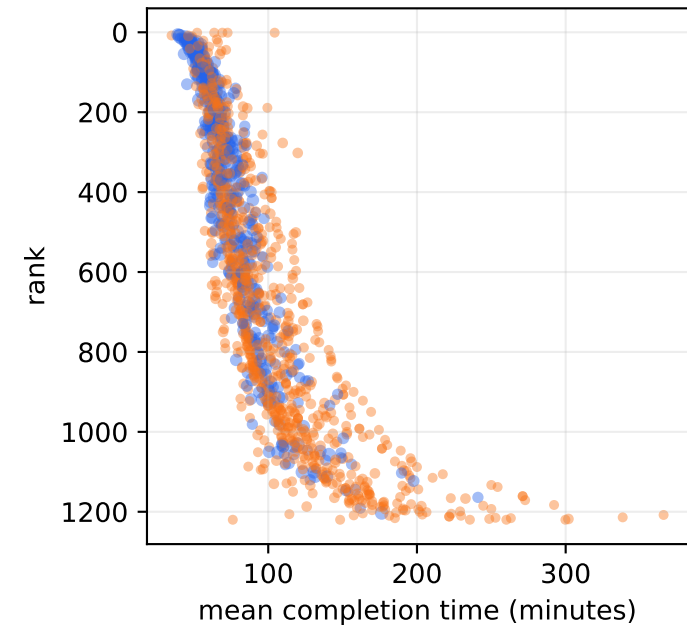
mean_event_percentile rank vs mean time
 $\rho=0.91$; orange=<3 events



weighted_normalized_rank rank vs mean time
 $\rho=0.88$; orange=<3 events

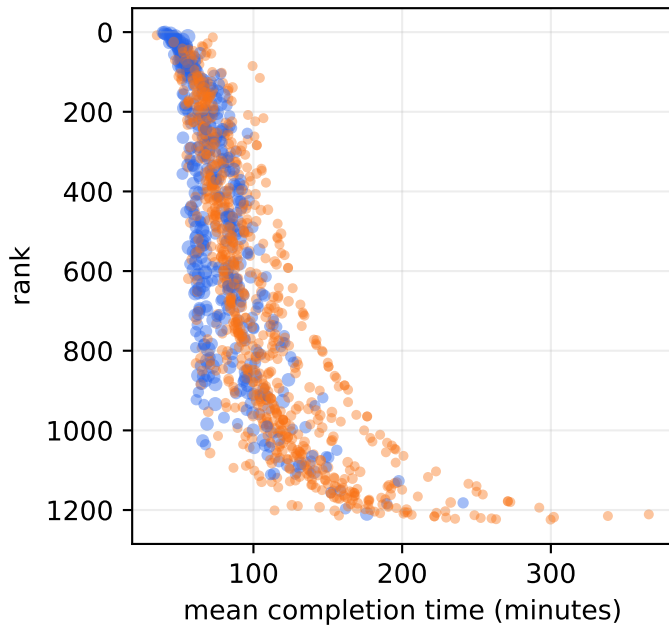


mean_normalized_rank rank vs mean time
 $\rho=0.90$; orange=<3 events

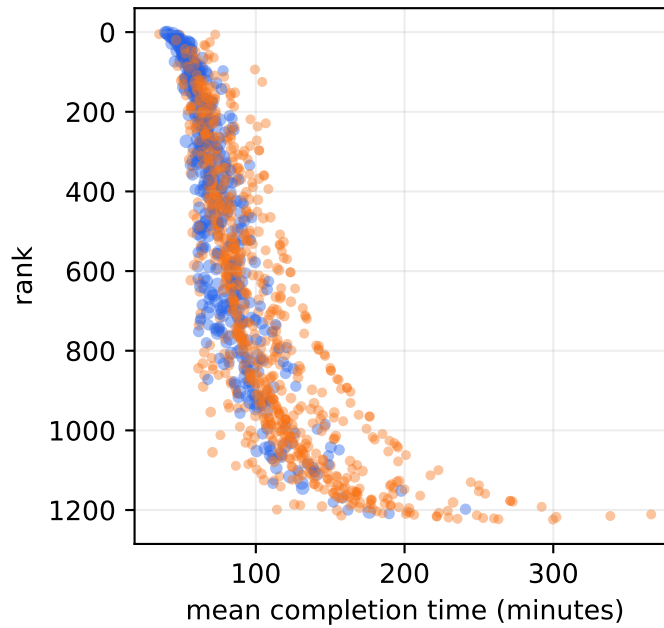


Rank vs Mean Time by Person (3/3)

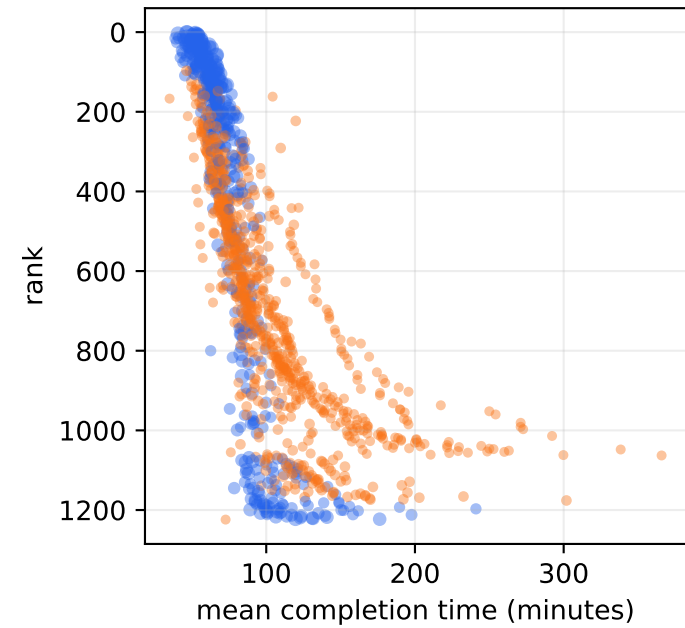
weighted_zscore rank vs mean time
 $\rho=0.82$; orange=<3 events



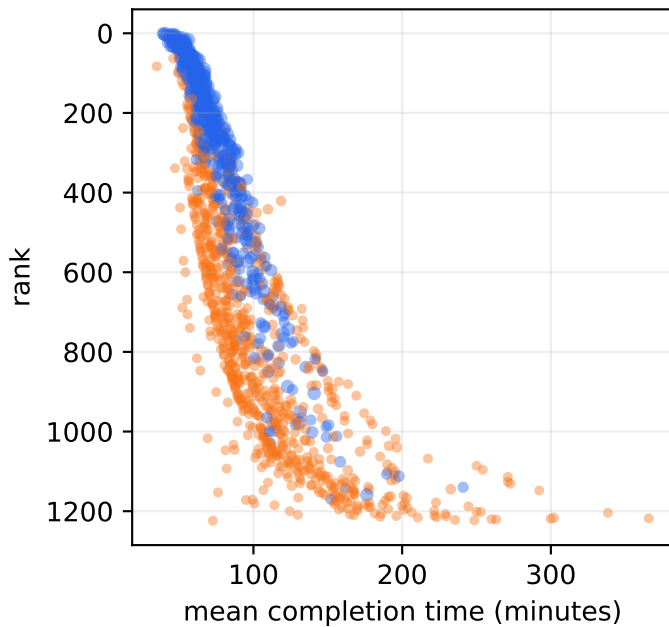
mean_zscore rank vs mean time
 $\rho=0.87$; orange=<3 events



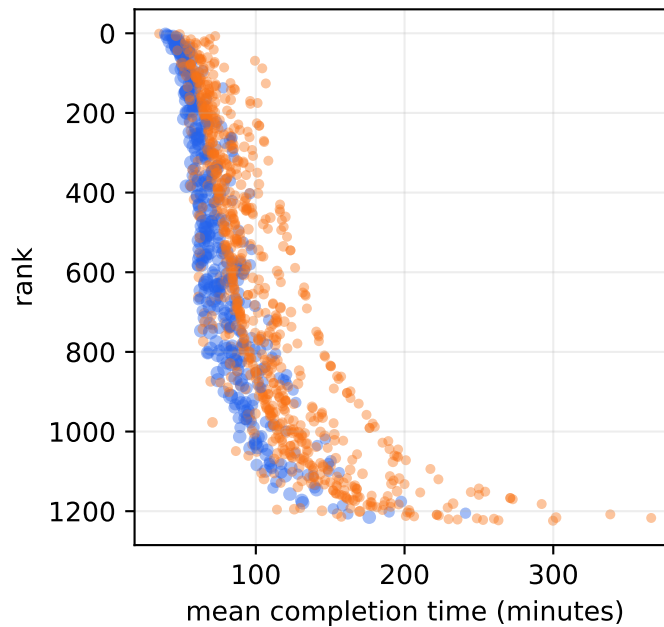
elo_rating rank vs mean time
 $\rho=0.87$; orange=<3 events



trueskill_conservative rank vs mean time
 $\rho=0.87$; orange=<3 events

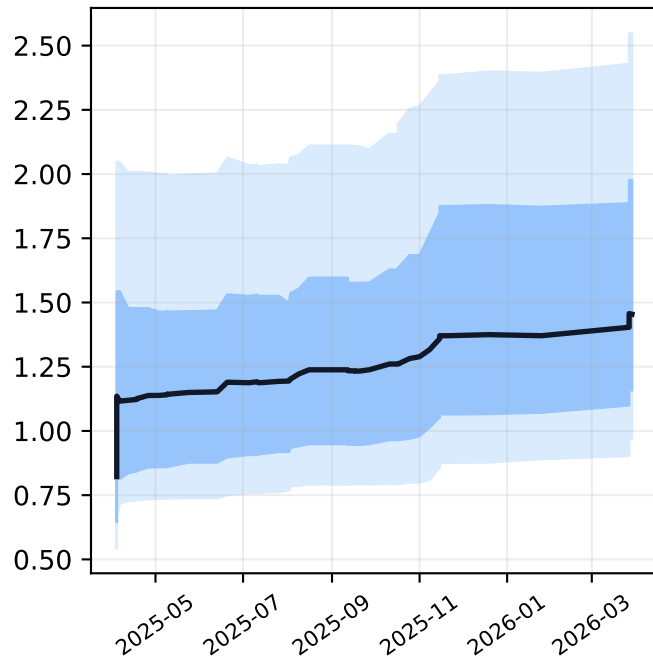


mss_like_score rank vs mean time
 $\rho=0.85$; orange=<3 events

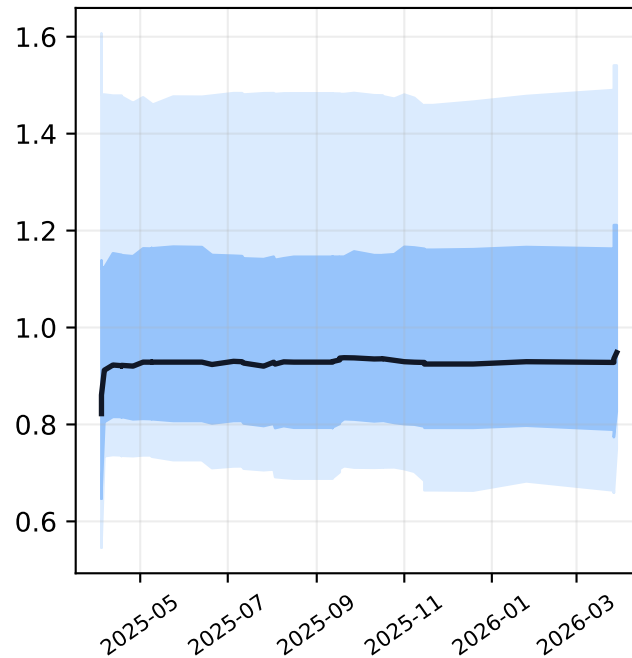


As-Of Distribution Drift (lower is better; Elo/TrueSkill negated)

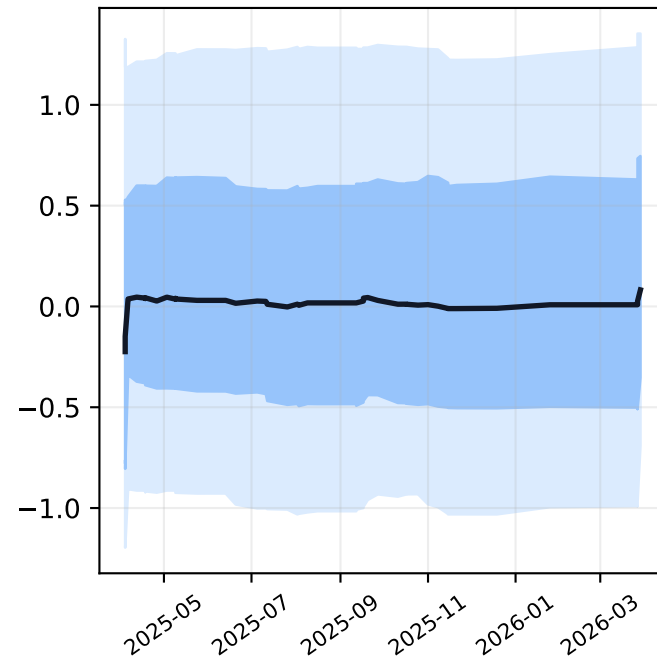
JPAR



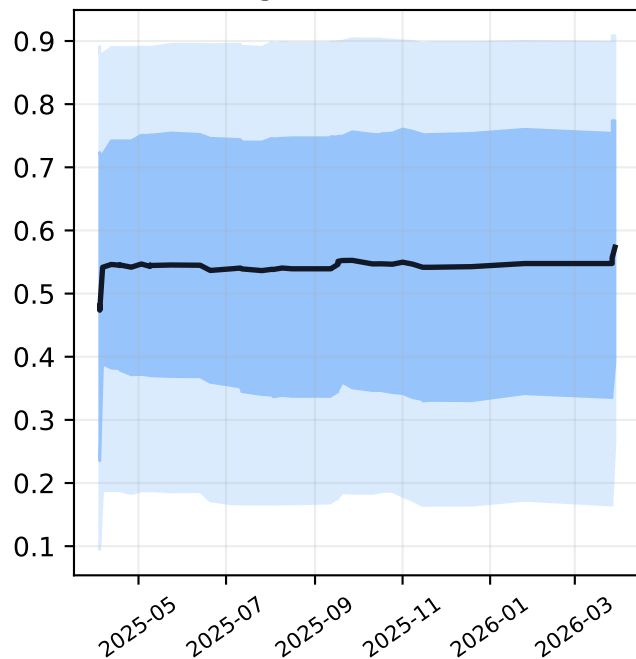
Raw JPAR Update



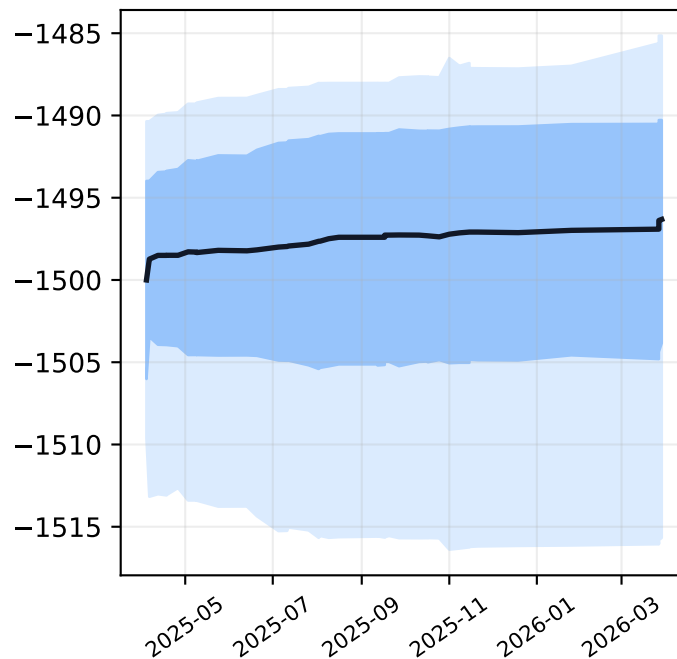
Weighted Log Z



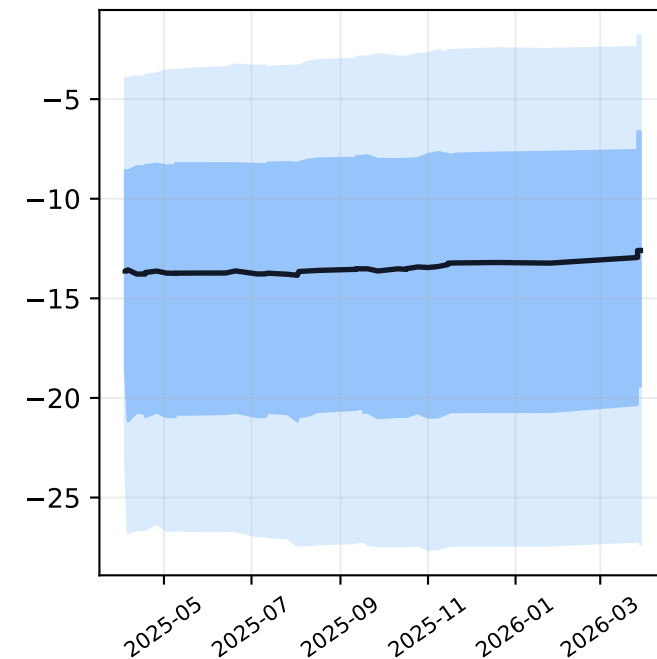
Weighted Percentile



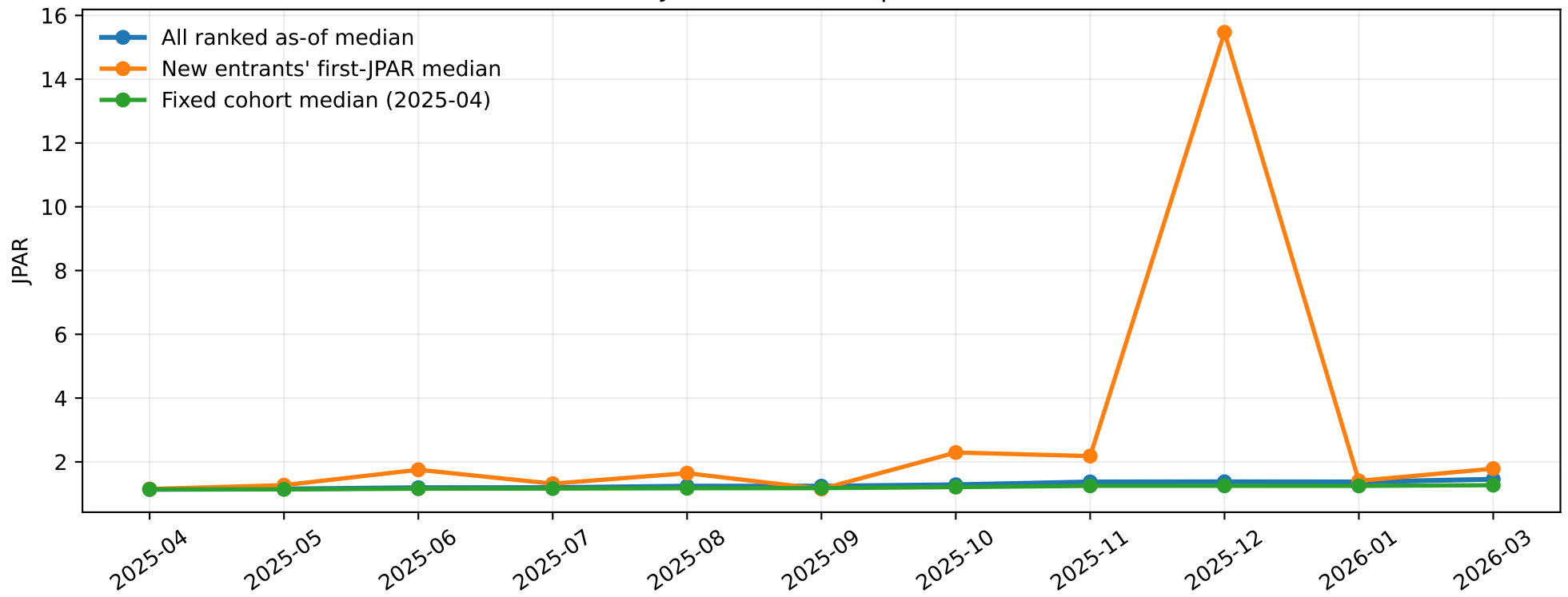
Elo



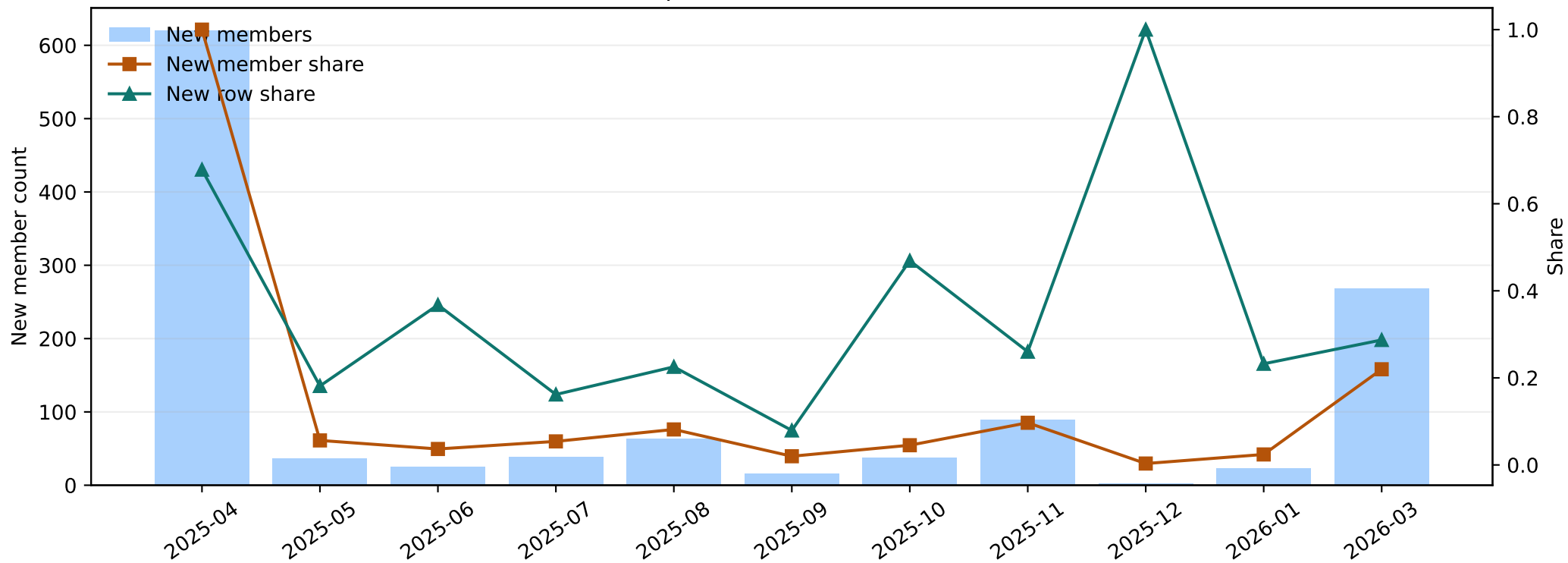
TrueSkill



JPAR Drift Decomposition

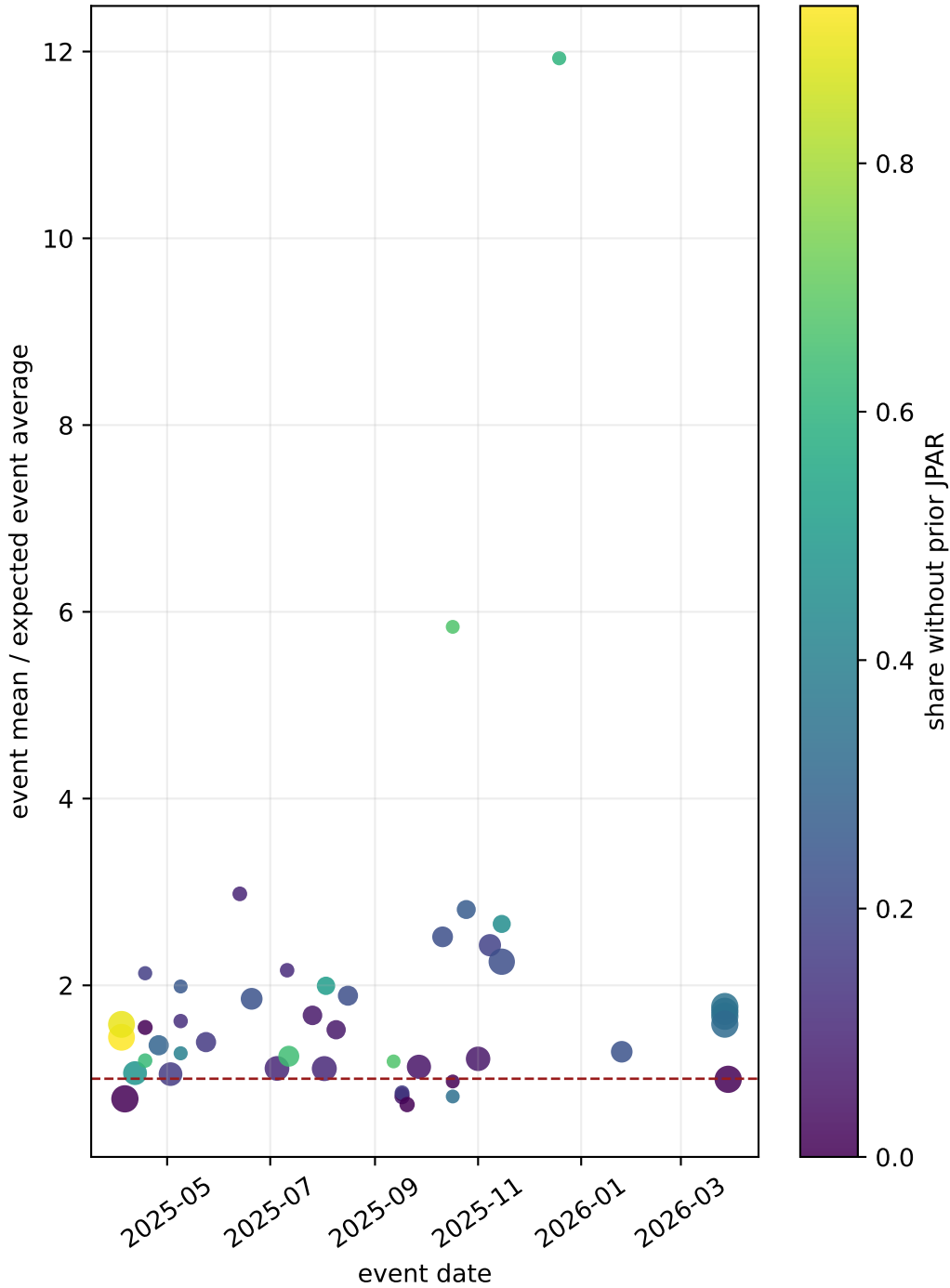


Population Mix Over Time

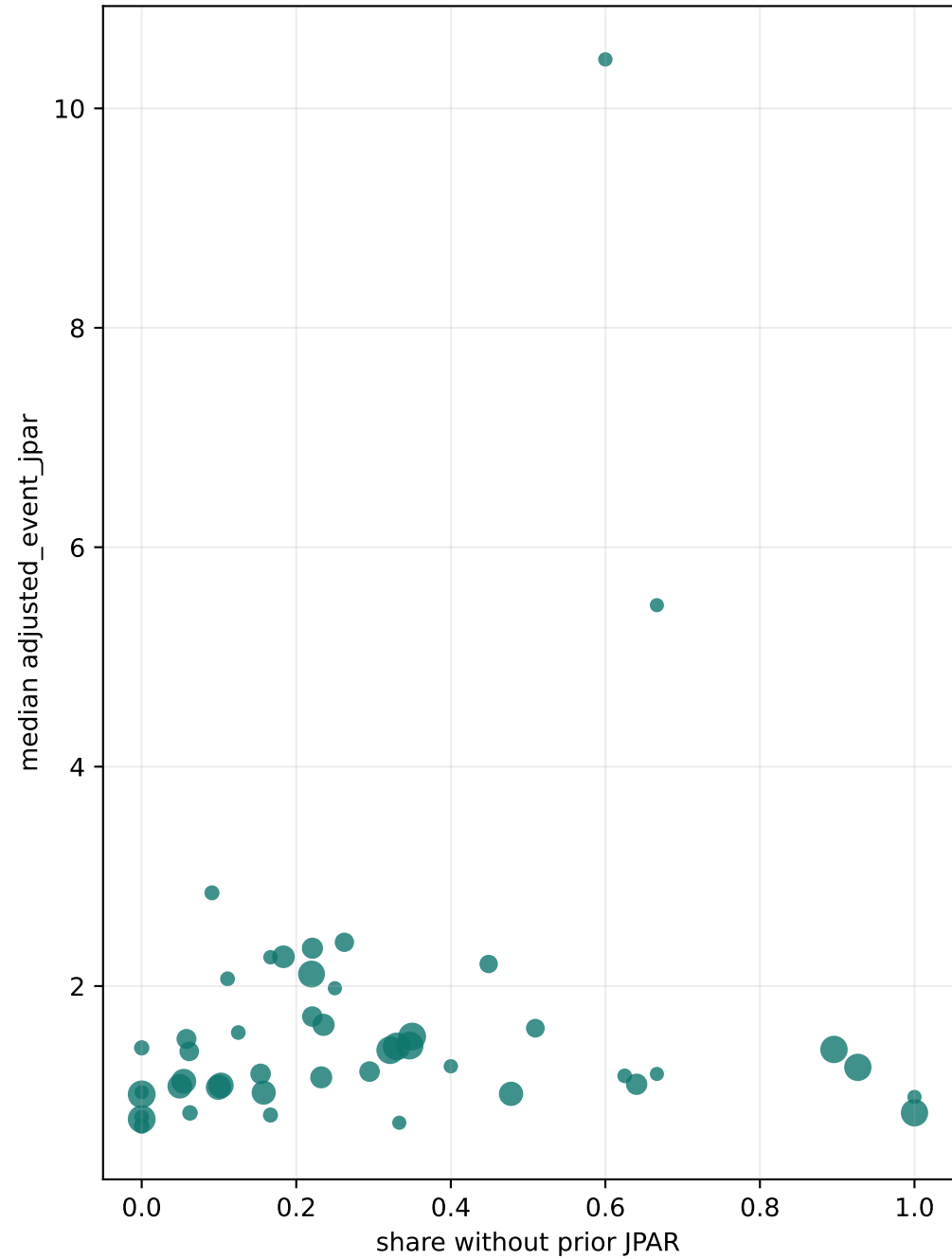


Event-Level Calibration Diagnostics

JPAR Calibration Multiplier



New Entrant Share vs Adjusted Event JPAR



Largest Rank Disagreements vs JPAR

system	full_name	events	jpar_rank	system_rank	rank_delta_vs_jpar	jpar
mvp_like_score	Mikayla Taylor	1	1222	25	-1197.0	6.631043
weighted_log_zscore	Mikayla Taylor	1	1222	47	-1175.0	6.631043
conservative_log_zscore	Mikayla Taylor	1	1222	73	-1149.0	6.631043
weighted_event_percent	Mikayla Taylor	1	1222	86	-1136.0	6.631043
mvp_like_score	Vicky Vetter	1	1219	88	-1131.0	5.348731
mvp_like_score	Kristi Frohreich	1	1221	126	-1095.0	5.471788
weighted_log_zscore	Vicky Vetter	1	1219	133	-1086.0	5.348731
elo_rating	Mikayla Taylor	1	1222	162	-1060.0	6.631043
elo_rating	Vicky Vetter	1	1219	162	-1057.0	5.348731
mvp_like_score	Debby Kilgore	1	1085	69	-1016.0	2.462093
conservative_log_zscore	Vicky Vetter	1	1219	205	-1014.0	5.348731
weighted_event_percent	Vicky Vetter	1	1219	210	-1009.0	5.348731
weighted_log_zscore	Kristi Frohreich	1	1221	226	-995.0	5.471788
weighted_log_zscore	Debby Kilgore	1	1085	108	-977.0	2.462093
conservative_log_zscore	Kristi Frohreich	1	1221	301	-920.0	5.471788
conservative_log_zscore	Debby Kilgore	1	1085	167	-918.0	2.462093
mvp_like_score	Olivia Oliphant	1	979	84	-895.0	2.150489
mvp_like_score	Michelle Walker	1	968	74	-894.0	2.119787
mvp_like_score	Patrick Madigan	1	997	107	-890.0	2.218171
weighted_event_percent	Katelyn Squire	1	919	33	-886.0	1.978143
mvp_like_score	Melissa Revelle	1	1018	135	-883.0	2.270503
mvp_like_score	Katelyn Squire	1	919	43	-876.0	1.978143
weighted_event_percent	Michelle Walker	1	968	93	-875.0	2.119787
mvp_like_score	Kelli Ryan	1	1033	160	-873.0	2.316555
mvp_like_score	Marenda Shurts	1	1038	168	-870.0	2.332603